ETH zürich

ETH Library

# Can a troubleshooting AI assistant improve task performance in industrial contexts?

**Journal Article**

**Author(s):**
Löwhagen, Nils; Schwendener, Philippe; Netland, Torbjörn

**Publication date:**
2025

**Permanent link:**
https://doi.org/https://doi.org/10.3929/ethz-b-000745594

**Rights / license:**
Creative Commons Attribution 4.0 International

**Originally published in:**
International Journal of Production Research, https://doi.org/10.1080/00207543.2025.2527368

This page was generated automatically upon download from the ETH Zurich Research Collection.
For more information, please consult the Terms of use.

INTERNATIONAL JOURNAL OF PRODUCTION RESEARCH
https://doi.org/10.1080/00207543.2025.2527368

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

# Can a troubleshooting AI assistant improve task performance in industrial contexts?

Nils Löwhagen, Philippe Schwendener and Torbjørn Netland

Department of Management, Technology, and Economics, ETH Zurich, Zurich, Switzerland

**ABSTRACT**

Access to domain expertise is critical to problem-solving activities. This study investigates the role and usefulness of an AI-based troubleshooting assistant in the knowledge-intensive context of train commissioning. The authors developed a multilingual, user-centred chatbot using a Retrieval-Augmented Generation framework and a Large Language Model to provide real-time, project-specific troubleshooting support. A controlled field experiment with 19 commissioning technicians completing 173 tasks was conducted to evaluate the effect of the AI assistant. Results show that AI-assisted users significantly outperformed non-users in task performance. The benefits were more substantial among less experienced technicians, which emphasises the prospect of AI assistants in bridging skill gaps. Performance gains were also moderated by the AI attitudes and AI familiarity of technicians, suggesting that organisations should take these factors into account when striving to adopt AI assistants. The findings provide empirical evidence for the role of generative AI in scaling operational knowledge, enhancing worker performance, and improving the efficiency of engineering workflows. This study contributes to the discussion of the role of generative AI in problem-solving by demonstrating a novel application for frontline decision support in complex, high-variance industrial tasks.

**ARTICLE HISTORY**
Received 15 November 2024
Accepted 28 May 2025

**KEYWORDS**
Generative AI; chatbot; knowledge management; retrieval-augmented generation; large language model; train commissioning

## 1. Introduction

Complex engineering projects, such as the commissioning process in train manufacturing, rely heavily on the specialised knowledge of skilled technicians. When a voltage discrepancy is detected in a train wagon – for example, a reading of 19 volts instead of the expected 36 volts – correctly identifying the cause requires domain-specific experience. Experienced commissioning engineers can apply their expertise to quickly diagnose such issues as wiring faults, thereby saving significant time and resources during troubleshooting. This example illustrates the critical role of immediate access to specialised knowledge in effective problem-solving. Consequently, an important question arises: how can less experienced commissioning engineers, who may be unfamiliar with such errors, be effectively supported? Addressing this challenge, advances in Artificial Intelligence (AI) offer promising opportunities for knowledge augmentation and decision support.

Recent studies suggest that Large Language Models (LLMs), a branch of generative AI, can facilitate such knowledge-sharing by augmenting decision-making and bridging experience gaps in knowledge-intensive industrial environments (Feuerriegel et al. 2024; Kernan Freire et al. 2024; Gröger, Schwarz, and Mitschang 2014). LLMs and other AI-enabled digital assistants can improve productivity and enhance task execution by providing real-time access to relevant information (Brynjolfsson, Li, and Raymond 2025; Zheng et al. 2024). Early investigations in operations and supply chain management have yielded promising results. For example, Jackson et al. (2024) show how the core capabilities of generative AI, such as prediction and adaptation, can enhance decision support, and Fosso Wamba et al. (2024) find that initial implementations of tools like ChatGPT have shown promising impacts on operational efficiency, cost reduction, and customer satisfaction.

Our study contributes to this nascent stream of literature by investigating if and how an AI-enabled troubleshooting assistant can improve productivity in a highly specialised engineering context: the commissioning of trains. Additionally, we examined whether workers' attitudes toward AI moderated the impact of the assistant on their performance. The study first reviews the extant literature to identify the requirements for an effective AI assistant. This includes examining essential

**CONTACT** Nils Löwhagen nloewhagen@ethz.ch Weinbergstrasse 56/58, 8092 Zurich, Switzerland

This article has been corrected with minor changes. These changes do not impact the academic content of the article.

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

knowledge areas and best practices for structuring information. This is followed by the design and implementation of the assistant prototype, integrating user-centred design elements to optimise usability in the commissioning context (Casillo et al. 2020; Hoerner, Schamberger, and Bodendorf 2023). The research team developed the AI-based assistant using a Retrieval-Augmented Generation (RAG) chain combined with an LLM, allowing technicians to access project-specific data and receive precise, up-to-date responses during the troubleshooting process.

The performance of the troubleshooting assistant is evaluated in a controlled field setting with 19 participants conducting a total of 173 realistic commissioning tasks to assess its accuracy, usability, and effectiveness in resolving common train commissioning issues. Survey and experimental data were collected to evaluate user satisfaction, efficiency metrics, and interaction feedback during realistic commissioning tasks with and without the support of the troubleshooting assistant. In particular, the analysis examines whether AI assistants have a stronger impact on less experienced engineers compared to their more experienced counterparts.

Traditional troubleshooting in train commissioning relies heavily on experienced technicians, creating knowledge bottlenecks that hinder efficiency and scalability. While AI-driven knowledge management has been explored in industrial contexts (Casillo et al. 2020; Ciancio et al. 2024; Farooqui et al. 2020; Kernan Freire, Wang, and Niforatos 2024; Gröger, Schwarz, and Mitschang 2014; von Wolff, Hobert, and Schumann 2019), context-aware, task-specific troubleshooting remains underexplored in the literature. The study advances the field by introducing a RAG-enhanced LLM assistant designed for real-time problem-solving in train commissioning. Our research enables technicians to retrieve and contextualise engineering knowledge efficiently, thereby mitigating dependencies on senior experts. Through empirical evaluation of its impact on task performance and knowledge dissemination, this study presents one of the first controlled experiments on generative AI for industrial troubleshooting. The findings offer important insights into the scalability of generative AI applications in manufacturing contexts.

The results demonstrated that the AI-enabled troubleshooting assistant significantly improved task performance, with notable benefits in answer quality, especially for less experienced participants and those with a favourable AI attitude. The results also indicated that AI familiarity is a decisive factor influencing the effectiveness of such tools in shop floor operations, highlighting the need for AI literacy training. In addition to answer accuracy, the ease of searching documentation and assistant usability were among the most valued features of the assistant. These findings emphasise the potential of LLMs to support technicians in completing complex tasks more effectively, especially those who may benefit from structured guidance.

The paper is organised as follows: Section 2 reviews prior research on AI-driven knowledge management and digital assistants. The design criteria, technological implementation, and user interface of the troubleshooting assistant are presented in Section 3. Section 4 describes the experimental research design used to evaluate the performance effects of the assistant. Section 5 evaluates the assistant's performance regarding user satisfaction and efficiency outcomes. Section 6 discusses the results in terms of novelty and contributions and offers implications for scholars and practitioners. Section 7 concludes the paper, discusses its limitations, and provides an outlook for future development.

## 2. Literature review

AI is increasingly integrated into all aspects of knowledge management, from knowledge creation and storage to retrieval and sharing across organisational departments. LLMs, in particular, offer flexible, human-like interactions that can outperform traditional intent-based systems in knowledge-intensive environments (Kernan Freire, Wang, and Niforatos 2024). For instance, Jarrahi et al. (2023) show that AI can *enhance* the creation and dissemination of expert knowledge across organisational silos, promoting collaboration and overcoming information barriers. AI systems can also contribute to knowledge management by *reducing* the dependence on domain experts, for instance through real-time anomaly detection and advanced problem-solving support (Kassner et al. 2017; Senoner, Netland, and Feuerriegel 2021).

### 2.1. AI assistants for knowledge management in manufacturing

In manufacturing, effective knowledge-sharing is essential for problem-solving, decision-making, and overall organisational performance (Kernan Freire et al. 2024). However, capturing and sharing tacit knowledge remains a key challenge (Johnson et al. 2019). Traditional knowledge management systems, in contrast, prioritise *know-what* and *know-how* information about products and processes because such information is codifiable. To date, they have been less effective for *know-why* knowledge, which tends to be more tacit and difficult to articulate and disseminate. In addition, knowledge repositories are usually passive, requiring users to search, interpret, and apply knowledge manually. As a result, problem-solving and troubleshooting remain challenging areas for knowledge

management, forcing manufacturers to rely on human experts for these tasks.

Domain knowledge, often held by experienced workers, is critical for managing complex production issues such as troubleshooting. However, experts are scarce and highly sought after, which limits the immediate access to expertise on the shop floor during troubleshooting (Gröger, Schwarz, and Mitschang 2014; Netland et al. 2023). Prior studies have explored various approaches to capturing and leveraging domain-specific knowledge. Hoerner, Schamberger, and Bodendorf (2023) propose a structured, mixed-method approach, while Kernan Freire et al. (2023) highlight the role of intelligent assistants with conversational interfaces in sharing tacit expertise. Similarly, Penica et al. (2023) present a decision-making chatbot integrated with Industrial Internet of Things (IIoT) devices and Augmented Reality (AR) interfaces to facilitate real-time information access, enabling operators to address complex tasks more efficiently. These studies emphasise the potential of AI, particularly LLMs, to enhance knowledge management systems. Such systems can reduce dependency on human experts by enabling fast, context-specific knowledge-sharing and problem resolution.

The integration of digital assistants, including chatbots, voice-activated systems, and virtual agents, is increasingly common in industrial environments (Colabianchi, Tedeschi, and Costantino 2023; von Wolff, Hobert, and Schumann 2019; Wellsandt et al. 2021), where they enhance real-time knowledge access and facilitate complex task execution. The multimodal interaction capabilities of digital assistants provide flexibility and adaptability, enabling them to meet diverse task-specific requirements. In supply chain contexts, generative AI tools have been implemented to support inventory management and demand forecasting, with early applications demonstrating reductions in processing time and error rates (Fosso Wamba et al. 2024). Zimmer et al. (2020) developed a decision-support chatbot that assists operators with equipment adjustments, reducing ramp-up time and minimising dependency on individual expertise. Similarly, Casillo et al. (2020) introduced a chatbot for employee training, significantly improving knowledge transfer and skill development among new employees. These implementations highlight the potential of generative AI to enable knowledge-sharing and improve workers' access to expert information in industrial contexts.

Hands-free, voice-based systems are particularly useful for production tasks requiring continuous physical engagement, while text-based interfaces are better suited for office tasks. By providing step-by-step guidance and troubleshooting support, such systems reduce dependence on experts and foster greater worker autonomy (Zheng et al. 2024). For example, Wellsandt et al. (2021) examine AI maintenance assistants that enable hands-free operations, automate routine maintenance tasks, and bridge skill gaps among personnel. Dolgui and Ivanov (2025) propose advancing digital assistants by integrating real-time behavioural insights, enabling more adaptive and context-aware support tailored to the dynamic needs of human workers. Providing targeted support to employees with varying experience levels is particularly valuable in contexts where specialised expertise is scarce (Zheng et al. 2024).

### 2.2. Technical LLM implementation of AI assistants

Recent technological advancements enhance the effectiveness of LLMs for industrial knowledge management. Techniques such as chain-of-thought prompting and RAG have proven particularly valuable in complex industrial settings (Tianyu Gao, Fisch, and Chen 2021; Lewis et al. 2020; Wei et al. 2023). Additionally, Virtual Reality (VR)-enabled chatbots for engineering consultations combine VR with Natural Language Processing (NLP) systems to support domain-specific knowledge-sharing and address critical expertise gaps (Trappey et al. 2022). Collectively, these advancements highlight the transformative potential of AI in improving knowledge management practices. They also help make *know-why* knowledge more accessible and tailored to the specific needs of industrial environments.

Integrating LLMs into industrial settings requires a targeted approach to training and fine-tuning, as general-purpose models such as ChatGPT (OpenAI) and Gemini (Google) often lack the technical specificity needed for specialised applications. Without adjustments, these models often struggle with domain-specific tasks that require in-depth technical knowledge, emphasising the need for tailored solutions. To address this, Yang et al. (2023) propose a hybrid setup where a smaller, fine-tuned LLM, trained on technical documentation, advises a more robust general LLM in question-answering tasks. While this configuration significantly enhances response relevance and accuracy, it is currently applied to specific domains, such as Microsoft Azure products (Yang et al. 2023). Similarly, Ouyang et al. (2022) demonstrate improvements in LLM alignment with user intent through human feedback, as exemplified in the InstructGPT models. This approach emphasises fine-tuning models to align with user preferences rather than increasing model size, thereby improving performance in instruction-following tasks without adding complexity (Ouyang et al. 2022).

Complementary techniques enhance LLM applicability in industrial environments by improving contextual understanding, especially in dynamic and detail-oriented tasks. For instance, Zalake and Naik (2019) utilise attention mechanisms within a chatbot model based on bidirectional Long Short-Term Memory (LSTM) networks structured in an encoder-decoder framework. This design allows the model to focus on relevant details, generating coherent responses even in complex environments, such as troubleshooting applications. These findings accentuate the importance of adapting generative models to meet the specific information requirements of industrial settings. Additionally, Wang et al. (2023) demonstrate that continuous user feedback and familiarity with voice-based AI systems enhance performance in complex customer service tasks, highlighting the value of sustained adaptation to enhance model performance in handling intricate, conversational queries in industrial contexts.

Further advancements emphasise innovative training methods to boost LLM reasoning capabilities. For example, Wei et al. (2023) introduce chain-of-thought prompting, where models are shown intermediate reasoning steps to enhance interpretability and problem-solving accuracy. Expanding on this, Lewis et al. (2020) propose RAG models that incorporate external knowledge sources, yielding factual and context-aware responses. Tianyu Gao, Fisch, and Chen (2021) introduce Language Model–Better Few-Shot Fine-Tuning (LM-BFF), a few-shot learning technique tailored for pre-trained models like Bidirectional Encoder Representations from Transformers (BERT). This method uses automated prompt generation to improve task performance with minimal data. These strategies collectively highlight the importance of combining specialised training, user-aligned feedback, and advanced prompting techniques to ensure LLMs can provide accurate, relevant, and reliable responses tailored to the complexities of industrial applications (Tianyu Gao, Fisch, and Chen 2021; Lewis et al. 2020; Ouyang et al. 2022; Wei et al. 2023; Yang et al. 2023).

### 2.3. Effects of AI assistants on performance

LLMs have demonstrated their effectiveness across diverse, knowledge-intensive fields by supporting decision-making and streamlining workflows. Whether applied in manufacturing, customer service, or legal analysis, LLMs enhance task-specific efficiencies while adapting to the distinct requirements of each sector (Choi and Schwarcz 2023; Kernan Freire, Foosherian, et al. 2023; Wang et al. 2023). By leveraging these adaptable capabilities, organisations can better manage knowledge and optimise workflows to complement human expertise across professional contexts.

LLM-based systems often outperform traditional intent-based models in knowledge-intensive settings, especially in manufacturing, enhancing response accuracy, reliability, and conversational fluidity. These improvements bolster productivity and support decision-making by providing precise, accessible information to users. For instance, models like GPT-3.5 are particularly effective at bridging knowledge gaps and increasing worker productivity by streamlining information retrieval and assisting task execution. In this context, Kernan Freire et al. (2024) and Kernan Freire, Wang, and Niforatos (2024) highlight how LLMs surpass traditional systems, while Kernan Freire, Foosherian, et al. (2023) outline specific implementation guidelines for LLMs in manufacturing. These guidelines stress the importance of domain-specific memory, real-time data integration, and robust data security. They help LLMs function as practical knowledge management tools within complex industrial contexts.

In addition to enhancing knowledge management, AI has yielded notable productivity gains across diverse operational contexts. For example, Brynjolfsson, Li, and Raymond (2025) report a 14% increase in issue resolution rates for customer support tasks through generative AI tools, significantly benefiting novice workers by accelerating skill acquisition and reducing experience gaps. Similar productivity improvements have been documented for knowledge-intensive tasks in other sectors. For instance, Dell'Acqua et al. (2023) found that consultants using generative AI completed tasks more quickly and with higher quality than those without AI support, emphasising the potential of task-specific AI integration to boost productivity and reduce disruptions.

Within manufacturing, generative AI applications have been shown to improve workflows and enhance information accessibility. For example, Do and Jeong (2022) developed ChatMES, a Robotic Process Automation (RPA)-powered chatbot that interacts with Manufacturing Execution Systems (MES), automating routine tasks and simplifying data access for technicians. This automation streamlines interactions with complex MES interfaces, saving valuable time. In semiconductor manufacturing, Loh et al. (2023) developed the Overall Equipment Efficiency AI (OEEAI) bot, which assists engineers by accurately identifying equipment inefficiencies, supporting continuous improvement, and reducing time spent on repetitive tasks. These implementations illustrate how AI-driven tools enhance productivity by reducing routine task time, streamlining workflows,

and facilitating efficient information flow in industrial settings.

LLMs also demonstrate adaptability and value across other sectors, such as customer service and legal analysis, where they address unique challenges. In customer service, for instance, Wang et al. (2023) show that voice-based AI systems can reduce customer complaints and improve service quality, particularly for straightforward requests. However, their findings indicate that users with prior experience interacting with AI systems are better equipped to handle complex queries. This reflects a learning curve in which familiarity with AI tools improves users' performance in handling complex interactions. In the legal domain, Choi and Schwarcz (2023) examine the impact of ChatGPT (GPT-4) on the performance of law students, noting that while the model assists with straightforward questions, its benefits diminish in complex tasks, particularly for advanced users. This highlights the need for thoughtful integration of AI to support rather than replace human expertise in fields requiring nuanced understanding.

### 2.4. Challenges and risks with AI assistants

While LLMs hold considerable potential to enhance productivity and decision-making, their integration into organisational workflows requires careful oversight to manage associated risks (Choi and Schwarcz 2023; Kernan Freire, Foosherian, et al. 2023; Wang et al. 2023). A primary concern is hallucinated responses – outputs that appear plausible but are ultimately incorrect – emphasising the need for continuous evaluation in real-world applications. This issue is amplified by biases and inaccuracies due to LLMs' reliance on vast, unfiltered datasets, as noted by Kernan Freire, Wang, and Niforatos (2024) and Brown et al. (2020). Feuerriegel et al. (2024) highlight the risks generative AI models present in operational environments, stressing the need for vigilant oversight, particularly in high-stakes applications where errors could have serious repercussions. To mitigate these risks, techniques such as RAG and knowledge graph integration improve factual consistency and reduce erroneous outputs by grounding responses in verifiable sources (Gilbert, Kather, and Hogan 2024). Additionally, the computational demands of LLMs and concerns about their potential misuse in misinformation campaigns further show the need for robust accountability mechanisms to ensure responsible deployment and regulatory compliance.

Beyond these technical risks, integrating LLMs poses substantial organisational and ethical challenges concerning data security, privacy, and human-centred job designs. As Choi et al. (2022) argue, a human-centric approach to AI is crucial, especially when handling sensitive data, to ensure that AI supports rather than replaces human expertise. Similarly, Ouyang et al. (2022) and Kernan Freire, Foosherian, et al. (2023) note that even with human feedback, models like InstructGPT may deviate from user intent or inadvertently compromise privacy, raising concerns about intellectual property and operational reliability. These insights advocate aligning AI systems with ethical guidelines and stringent data protection practices to mitigate potential harm.

Another significant challenge is the risk of over-reliance on AI, which can diminish task quality and human decision-making capabilities. Dell'Acqua et al. (2023) found that professionals who heavily depended on AI tools experienced increased error rates, emphasising the need to calibrate AI support to task complexity. To mitigate this risk, Jarrahi et al. (2023) advocate for AI literacy and training programs, which would help users view AI as an augmentative tool that enhances, rather than substitutes, human skills. This concern extends to creative industries, where Berg, Raj, and Seamans (2023) caution that generative AI may unintentionally stifle human-driven innovation. They recommend that organisations adopt adaptive strategies to safeguard the unique value of human creativity while leveraging AI to enhance productivity.

### 2.5. Identified research gap

Despite advancements in AI-driven knowledge management and digital assistants, a significant research gap remains in developing domain-specific AI tools for real-time troubleshooting in complex industrial environments. Existing studies examine AI's role in knowledge management (Kernan Freire et al. 2024), decision support (Fosso Wamba et al. 2024), and production automation (Jackson et al. 2024), yet these applications lack the immediacy and context sensitivity required for effective troubleshooting. Unlike general AI-powered knowledge-sharing tools, troubleshooting demands precision, immediate applicability, and trustworthiness in high-stakes environments.

Standard LLM-based assistants provide broad, probabilistic responses that may not be sufficiently precise for technical troubleshooting (Kernan Freire et al. 2023). Manufacturing environments demand AI systems capable of producing technically accurate, context-aware responses aligned with operational constraints. Traditional knowledge management systems depend on passive search and retrieval, forcing technicians to manually interpret and apply information – an inefficient process under time constraints (Hoerner, Schamberger, and Bodendorf 2023). In troubleshooting, delayed or

inaccurate guidance can lead to operational disruptions, downtime, and financial losses. AI-driven troubleshooting tools must be reliable and transparent, as technicians must rely on AI-generated recommendations, unlike customer support or supply chain applications, where errors are more easily corrected (Ouyang et al. 2022).

An often-overlooked factor in AI-assisted troubleshooting is how industrial experience influences problem-solving. Lugovoi, Andritsos, and Senot (2022) highlight that while related experience improves efficiency, it may limit innovation, whereas unrelated experience fosters innovation but increases complexity. In troubleshooting, experienced workers may default to familiar solutions, while less experienced technicians struggle to contextualise AI-generated insights. Existing AI research does not adequately explore how AI can complement different levels of expertise in troubleshooting contexts.

Most AI-driven digital assistants in manufacturing, such as ChatMES (Do and Jeong 2022) and the OEEAI bot (Loh et al. 2023), focus on structured automation rather than real-time troubleshooting. AI adoption studies typically address long-term decision-making rather than immediate, task-specific interventions. For example, von Wolff, Hobert, and Schumann (2019) explore conversational AI in industrial settings but not its direct impact on troubleshooting workflows. Telner (2021) found that chatbots often fail to meet user expectations due to generic responses, requiring users to refine queries repeatedly to obtain useful information. Ambiguous or inaccurate AI-generated answers often foster user skepticism and reluctance to rely on chatbot guidance. Speed and content quality are key to user satisfaction, reinforcing the argument that AI assistants must prioritise precision over broad, probabilistic outputs.

Rai et al. (2021) emphasise that most AI applications in Industry 4.0 focus on predictive maintenance, process optimisation, and fault detection rather than real-time technician support. While these applications demonstrate the utility of AI in manufacturing, they lack the real-time decision support required for troubleshooting. Their study demonstrates the necessity of tailoring AI solutions to specific industrial contexts and highlights the importance of fine-tuned, domain-specific AI assistants.

This study addresses the gap by developing and empirically testing a RAG-integrated AI troubleshooting assistant for train commissioning, ensuring accuracy and context relevance. It examines the impact of AI on task performance and knowledge transfer, with a focus on different levels of user expertise, and investigates whether AI can bridge skill gaps in troubleshooting. By validating the AI assistant in real-world commissioning scenarios, this research contributes novel insights into the practical application of AI-driven troubleshooting assistants in industrial environments. Unlike studies centred on AI's theoretical potential for knowledge sharing, this study directly examines how AI enhances accuracy, efficiency, and accessibility in troubleshooting.

## 3. An AI-enabled troubleshooting assistant

This section introduces the research setting and describes the development of the AI troubleshooting assistant. It outlines the specific requirements for supporting the train commissioning process at the case company and details the assistant's technical implementation and user interface.

### 3.1. Research setting

This study was conducted in collaboration with TrainCo, a pseudonym for a global manufacturer of railway rolling stock with facilities across multiple countries. The research focuses on TrainCo's commissioning process, a critical step in deploying new rail vehicles. Train commissioning represents the final stage of manufacturing, during which all systems and safety features of a new train are thoroughly tested, validated, and certified. This ensures compliance with operational standards and readiness for passenger service. TrainCo's complex operations and emphasis on precision engineering provide an ideal setting to study the effects of generative AI support for expert tasks in manufacturing.

The commissioning process is divided into two parts: wagon commissioning and train commissioning. During wagon commissioning, the wagon's wiring system is inspected to verify its alignment with the wiring diagram. Train commissioning ensures the vehicle is fully operational and prepared for delivery to the customer. This is achieved by sequentially commissioning each subsystem installed on the train. Additionally, the functionality of the individual subsystems and the complete train is rigorously tested against predefined protocols. In both stages of commissioning, most procedures are carried out manually by a commissioning technician, with minimal automation. Consequently, the process depends heavily on the technician's training, expertise, and experience.

### 3.2. Technical implementation

We developed a prototype AI-enabled troubleshooting assistant to support technical documentation during the commissioning phase of a train manufacturing project. We continuously incorporated feedback from TrainCo's Commissioning Department to align the assistant's functionality with operational needs.

The primary objective of the troubleshooting assistant is to provide commissioning technicians with real-time access to specialised knowledge. Specifically, we aimed to facilitate knowledge transfer and reduce reliance on experienced employees, who traditionally train and support new staff. By handling routine inquiries, the assistant reduces the workload of senior personnel while equipping new employees with adequate support to perform their roles effectively.

In collaboration with TrainCo, we defined the following design requirements for the troubleshooting assistant through consultations with managers and commissioning technicians. (1) The assistant must support simultaneous access by multiple users across different locations, necessitating a web-based application. (2) The assistant should feature a simple, intuitive interface, implemented as a screen-based chatbot. (3) It must process documents and queries in multiple languages, requiring LLM integration. (4) Responses must be grounded in the most current TrainCo documentation; therefore, specific documents were selected as reference sources instead of the entire project database. (5) To ensure reliability, the assistant must include citations and direct links to the relevant paragraphs within source documents in its outputs.

### 3.2.1. Retrieval-augmented generation

As outlined in the literature review, a RAG chain is well-suited for handling user queries. Since this study focuses on developing and evaluating a prototype to support commissioning engineers, we selected the OpenAI API for all AI-related tasks. This choice is supported by Kernan Freire et al. (2024), who found that OpenAI's foundation model outperforms others in industrial contexts. RAG is an NLP technique that combines retrieval-based and generation-based approaches. The primary goal of this combination is to enhance the performance and accuracy of the language model. Another key advantage of RAG is its ability to reduce hallucinations in LLMs by grounding responses in structured knowledge sources (vector storages) rather than relying solely on the model's internal representations. For instance, Gilbert, Kather, and Hogan (2024) demonstrate that RAG reduces hallucinations compared to standalone LLMs lacking structured knowledge integration, which often results in factual errors and outdated information.

Figure 1 illustrates the three steps of the RAG chain, adapted from Yunfan Gao et al. (2023) with minor modifications. This process consists of three sequential phases: Retrieval, Augmentation, and Generation. In the Retrieval Phase, the system identifies and retrieves text segments or data snippets most relevant to the user's query. In the Augmentation Phase, the retrieved content is enriched with supplementary information to provide a more contextualised input for the generative AI model. This augmentation step is crucial for mitigating hallucinations, as it ensures that generated responses are both coherent and factually grounded in reliable data sources (Gilbert, Kather, and Hogan 2024). In this study, the augmentation process involves configuring the AI to assist field engineers with troubleshooting tasks. This setup includes assigning queries to an appropriate subsystem, distinguishing between multiple potential solutions when applicable, and responding in the language of the original query. Finally, in the Generation Phase, the AI uses the enriched input to formulate a response. The retrieved context and targeted augmentations guide the tool's output, ensuring precise and context-sensitive replies.

### 3.2.2. Architecture

The architecture of the AI-enabled troubleshooting assistant comprises five key components: *User*, *Document Management*, *Azure Cloud*, *Enterprise Network*, and *OpenAI*. In this study, the *User* refers to the commissioning technician who interacts with the troubleshooting assistant by posing queries. The *Document Management* module manages the selection of documents used by the assistant, from which relevant information chunks are retrieved during the Retrieval Phase. The *Azure Cloud* hosts the chatbot frontend and the document database, facilitating access to stored information. The *Enterprise Network* serves as the primary repository for documents referenced by the chatbot in its responses. Finally, *OpenAI* performs the AI-driven processes of the RAG chain via Application Programming Interface (API) access to its services. We used the LlamaIndex data framework[1] to power the query engine and construct the vector store index. The chatbot interface was developed using the Flask framework[2]. The following sections provide a detailed overview of these components and their interactions.

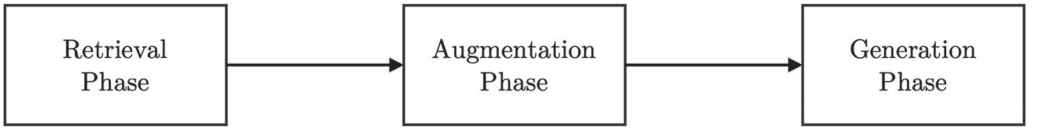



**Figure 1.** Workflow of the Retrieval-Augmented Generation chain.

#### 3.2.3. Importing documents into the document manager

The document manager manages the selection of documents to be processed by the AI-enabled troubleshooting assistant. Selected documents undergo a preprocessing sequence to prepare them for retrieval. Each document is initially divided into chunks of 1024 tokens, with a 20-token overlap. Metadata, such as the original document name and page number, is appended to each chunk to facilitate reference in responses to user queries.

Subsequently, these chunks are transformed into embeddings using the 'text-embedding-ada-002' model from the OpenAI API. This model was selected for its strong ability to capture semantic relationships while maintaining efficiency and scalability. Embeddings are dense vector representations that capture semantic information, representing semantically similar concepts with similar vectors. This approach ensures that embeddings are of fixed size and independent of vocabulary, enabling consistent representation.

The embeddings generated via the OpenAI API are stored in a MongoDB instance hosted in the Azure environment. Given the computational intensity of the embedding process, it is performed once per document. Re-embedding is only necessary when a document is updated, allowing the new content to be integrated into the embedding store.

### 3.3. Using the AI assistant during commissioning tasks

When a commissioning technician submits a query to the chatbot, the RAG process is initiated, executing the three main phases shown in Figure 1. First, the technician's query is sent to the OpenAI API, where it is transformed into an embedding vector. Embeddings with similar meanings have closely aligned vector representations. Therefore, standard vector similarity metrics are applied to identify the most relevant vectors in the MongoDB database. This database comprises documents relevant to the technician's query. During the Retrieval Phase, the application retrieves the 15 closest vector matches, though this parameter can be adjusted as needed.

The retrieved embeddings, referred to as context information, are combined with additional predefined data and the user's original query to form a single, enriched request. This consolidated input is then sent to the generative AI model, GPT-3.5-turbo from OpenAI, in a process known as the Augmentation Phase.

Each user query triggers two distinct database calls within the application. The first call accesses project documentation, including commissioning instructions, maintenance manuals, supplier system descriptions, and the operation and maintenance manual from the engineering department. The second call retrieves commissioning protocols, which are detailed records maintained by the commissioning department that document each step required for subsystem commissioning.

The final response consolidates the outputs from both database calls into a single message, clearly indicating the source of each answer segment. Each response section concludes with source references, including direct links to specific passages within the source documents. These links are generated using metadata from the 15 closest text chunks identified during the Retrieval Phase, including the source document and page number. This metadata enables the system to generate links directing users to the relevant paragraphs in the original documents.

## 4. Evaluation method

We conducted a controlled field experiment to evaluate whether the AI-enabled troubleshooting assistant improved the quality and efficiency of participants' responses to realistic project-related technical questions. The experimental design is illustrated in Figure 2.

### 4.1. Experimental design and study population

In a between-subjects experiment design, participants from TrainCo completed a pre-experiment survey to assess their perceptions of AI. Participants in the chatbot condition also completed a post-experiment survey to assess their experience using the chatbot. We recruited 19 qualified workers at TrainCo for the experiment. This number covers the entire available workforce in the commissioning department at the selected facility, as commissioning is a highly specialised task requiring extensive expertise. Including workers from other departments was considered but ultimately deemed inappropriate, as differences in expertise could complicate the interpretation of troubleshooting performance.

The controlled field experiment evaluated the chatbot's effectiveness in supporting technicians with realistic daily tasks. Participants completed a questionnaire developed in collaboration with the project manager, comprising 22 task-based scenarios linked to specific train subsystems, e.g. exterior doors, traction motors, and passenger information systems. This ensured task relevance and authenticity (Appendix 2). Participants were divided into two groups: one with chatbot assistance and a control group without chatbot assistance. This setup enabled a comparative assessment of task performance, providing a baseline for responses without AI support.

15 participants completed 143 tasks with chatbot assistance, while four participants completed 30 tasks without

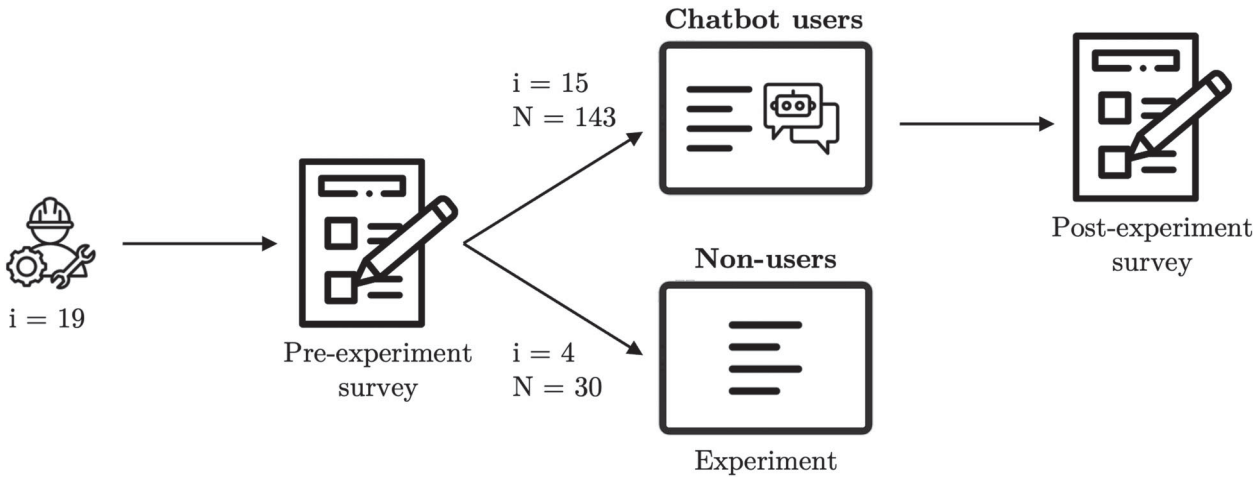



**Figure 2.** Overview of the experimental setup.

it, resulting in a total of N = 173 commissioning tasks. Given the limited participant pool, we opted for a larger sample for the chatbot-assisted condition to maximise opportunities for collecting user feedback on the AI assistant. Responses were rated on a five-point Likert scale (1 = very poor, 5 = excellent) based on a standardised sample solution. Non-responses were excluded to focus on answer quality. Performance was evaluated across four key metrics: total score, average score per task, task count (indicating response speed), and, for chatbot users, a subjective rating of chatbot responses on a ten-point scale (1 = very poor, 10 = excellent). Each participant had 45 minutes to answer as many troubleshooting tasks as possible. Participants completed an average of nine tasks each.

### 4.2. Pre-experiment survey

The pre-experiment survey, completed by all 19 participants, gathered data on their backgrounds, perceptions of AI, and workplace applications of AI. This information supported the development of two key constructs – *AI Efficiency* and *AI Attitude* – which were used to analyze the chatbot's effectiveness in this study. Survey details are provided in Appendix 1. Participants were employed in either commissioning (63%) or software engineering (37%). Educational backgrounds varied: 63% had completed an apprenticeship, while smaller proportions held bachelor's or master's degrees, 16% and 21% respectively. Professional experience levels ranged as follows: 16% had less than one year of experience, 32% had 1–3 years, 16% had 3–5 years, and 32% had more than 5 years of experience. This diversity in expertise and education served as a basis to explore whether these variables moderated the experimental results.

### 4.3. Outcome variables

As previously described, two constructs were developed to evaluate participants' perceptions and attitudes toward artificial intelligence: *AI Efficiency* and *AI Attitude*. Each construct is a composite variable derived from responses to selected Likert-scale items, capturing particpants' perceptions of AI's impact.

*AI Efficiency* assesses participants' perceptions of AI's impact on work efficiency. It is based on three survey items – Q14, Q15, and Q16 (Appendix 1) – in which participants rated the extent to which they believe AI enhances efficiency, adds interest to their tasks, and aids decision-making. Together, these items reflect perceived effectiveness in improving job performance. The construct was calculated as an unweighted average of responses to these items.

*AI Attitude* is derived from five items – Q7, Q8, Q9, Q10 (reverse-coded), and Q11 (Appendix 1) – capturing participants' views on the overall influence of AI, its potential for job creation, quality-of-life improvements, data protection concerns, and trustworthiness. We calculated the composite score for this construct as an unweighted average of responses to these items.

We calculated Cronbach's Alpha[3], to assess the internal consistency of each construct. The analysis yielded values of 0.84 for *AI Efficiency* and 0.79 for *AI Attitude*. These high scores indicate strong internal reliability, reflecting that the items within each construct measure cohesive underlying concepts. Thus, both constructs can be considered reliable measures for subsequent analyses.

Table 1 provides descriptive statistics for each construct, including the mean, standard deviation, minimum, and maximum values. The mean score for *AI Efficiency* suggests that participants generally perceive AI's impact on work efficiency as moderate, with considerable

**Table 1.** Descriptive statistics for AI constructs.

| Variable | N | Range | Mean | Std. dev. | Min | Max |
|---|---|---|---|---|---|---|
| *AI Efficiency* | 19 | 1–10 | 4.91 | 2.17 | 0.00 | 8.33 |
| *AI Attitude* | 19 | 1–10 | 4.44 | 1.47 | 2.20 | 8.20 |

variability indicated by the standard deviation. This variation reflects a diversity of opinions, from highly positive to more critical views on AI's role in enhancing efficiency. In contrast, the mean score for *AI Attitude* indicates a neutral to slightly negative perspective on AI's societal and technological impact. Variability in *AI Attitude* is less pronounced than in *AI Efficiency*, reflecting more consistent and balanced views among participants. These results highlight varied perceptions of AI's utility in the workplace, with *AI Efficiency* displaying broader variability and *AI Attitude* indicating a more consistent, neutral to slightly negative outlook.

We conducted a Pearson correlation analysis to assess the relationship between *AI Efficiency* and *AI Attitude*. The analysis revealed a moderate positive correlation ($r = 0.57$, $p = 0.01$), suggesting that participants perceiving AI as beneficial for enhancing work efficiency also tend to hold generally positive attitudes toward AI. This statistically significant result indicates that the observed association is unlikely to have occurred by chance. It supports a meaningful link between perceived work-related benefits of AI and favourable attitudes toward its broader societal and technological impact. As expected, this finding suggests that positive work-related experiences with AI contribute to more favourable overall attitudes toward its broader use.

## 5. Results

Table 2 presents an overview of participants' performance outcomes. Total scores, reflecting overall task performance, exhibited substantial variability with some participants excelling while others encountered difficulties. In contrast, average scores per task showed greater consistency, clustering around the midpoint. The broad range in task count reflected varying completion speeds and time constraints among participants. Chatbot users generally rated the assistant positively, with satisfaction scores ranging from moderate to high. These initial findings, characterised by substantial score variability, indicated significant differences in task performance and completion rates, alongside generally positive responses from chatbot users. These results provide the basis for subsequent comparisons between chatbot-assisted and non-assisted groups.

**Table 2.** Descriptive statistics for the performance measures.

| Variable | N | Range | Mean | Std. dev. | Min | Max |
|---|---|---|---|---|---|---|
| Total score | 19 | *n*–110 | 22.47 | 13.06 | 9.00 | 54.00 |
| Average score per task | 19 | 1–5 | 2.48 | 0.67 | 1.00 | 3.50 |
| Task count | 19 | 0–22 | 9.11 | 4.42 | 3.00 | 19.00 |
| Average chatbot rating | 15 | 1–10 | 6.36 | 1.00 | 4.20 | 7.83 |

*n* = the number of tasks the participant completed

### 5.1. Experimental results

In the first analysis, we compare the overall performance between the treatment group (chatbot-assisted) and the control group (no chatbot). To conduct this comparison, we used Mann-Whitney U tests to assess total scores, average scores per task, and task count. Given the small sample size, these non-parametric tests offered robust insights without assuming a normal distribution. The results showed significant differences for total scores ($p = 0.035$) and average scores per task ($p = 0.005$), with chatbot users outperforming non-users on both measures. These findings demonstrate that access to the chatbot led to better task response quality. However, task count showed no significant difference between groups ($p = 0.339$), indicating that while the chatbot enhanced response quality, it did not affect the number of tasks completed. Figure 3 illustrates these differences, displaying higher median scores and a wider performance range for chatbot users in total and average task scores. In contrast, task counts remained similar between groups.

In an additional test, we leveraged the repeated tasks per participant to analyze performance at the task level. We analyzed each completed task as a separate instance ($N = 173$) to examine differences between chatbot users and non-users. Of the 173 completed tasks, 143 were completed with the chatbot and 30 without. We first assessed the normality assumption of the task score distribution. A Skewness and Kurtosis test indicated significant deviation from normality ($\chi^2(2) = 12.250$, $p = 0.002$), primarily due to excess kurtosis ($p < 0.001$). Given the violation of normality, we conducted both parametric (t-test) and non-parametric (mixed-effects) analyses to ensure robustness.

We conducted a Welch's t-test, which does not assume equal variances. The results, shown in Table 3, revealed a significant difference in scores between chatbot users and non-users. Chatbot users consistently outperformed non-users.

However, since each of the 173 task responses was provided by one of 19 participants, the assumption of independent observations is likely violated. To account for this, we fitted a mixed-effects model with participant-level random intercepts. In this model, chatbot use was specified as a fixed effect, while participant ID served as

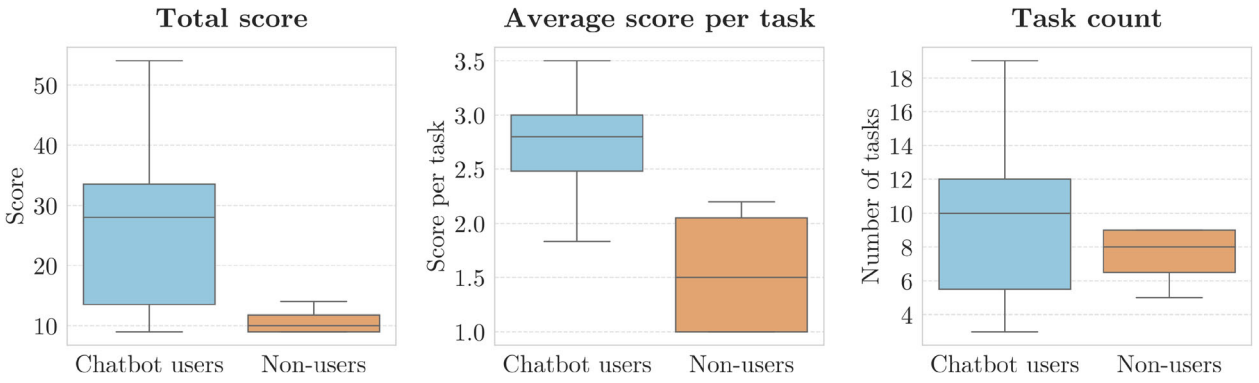



**Figure 3.** Performance comparison between experimental groups.

**Table 3.** Welch's t-test for independent samples using the full sample of 173 completed tasks.

| Group | Observations | Mean | Std. err. | Std. dev. | [95% CI lower] | [95% CI upper] |
|---|---|---|---|---|---|---|
| Non-users | 30 | 1.433 | 0.114 | 0.626 | 1.200 | 1.667 |
| Chatbot users | 143 | 2.685 | 0.088 | 1.058 | 2.510 | 2.860 |
| Combined | 173 | 2.468 | 0.084 | 1.103 | 2.303 | 2.634 |
| diff | | −1.252 | 0.145 | | −1.540 | −0.964 |

diff = mean(0) − mean (1), $t = -8.663$, Ha: diff $\neq 0$, $Pr(|T| > |t|) < 0.000$

**Table 4.** Mixed-effects model for task performance using the full sample of 173 completed tasks.

| Variable | Coefficient | Std. err. | [95% CI lower] | [95% CI upper] |
|---|---|---|---|---|
| Chatbot | 1.226*** | 0.245 | 0.746 | 1.706 |
| Intercept | 1.472*** | 0.220 | 1.041 | 1.903 |
| | Random effects | | | |
| Participant-level variance | 0.068 | 0.060 | 0.012 | 0.386 |
| Residual variance | 0.919 | 0.105 | 0.735 | 1.150 |

a random intercept to account for individual differences. The model provided evidence that chatbot use was a significant predictor of performance, with scores increasing by an average of 1.226 points. A likelihood ratio test suggested that while accounting for participant differences slightly improved the model score variance was predominanetly explained by task-level factors. This performance improvement can be attributed to the AI assistant, rather than individual participant differences. Table 4 presents the detailed results of the mixed-effects model.

### 5.2. Investigation of moderating variables

To further investigate factors affecting performance at the participant level, we conducted regression analyses using total score and average score per task as dependent variables. Independent variables included the pre-experiment constructs, chatbot usage, and demographics such as experience, education, and AI familiarity.

**Table 5.** Regression results for total score.

| Variable | Coefficient | Std. err. | [95% CI lower] | [95% CI upper] |
|---|---|---|---|---|
| *AI Efficiency* | −0.460 | 0.474 | −1.503 | 0.582 |
| *AI Attitude* | 2.505** | 0.934 | 0.449 | 4.561 |
| Experience | 1.853* | 0.933 | −0.200 | 3.907 |
| Education | 2.241* | 1.235 | −0.478 | 4.960 |
| AI familiarity | −1.859 | 1.298 | −4.716 | 0.998 |
| Chatbot usage | 7.277** | 2.515 | 1.741 | 12.813 |
| Task count | 2.114*** | 0.224 | 1.620 | 2.607 |
| Intercept | −16.764*** | 4.341 | −26.319 | −7.209 |

Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$. Model Fit Statistics: $R^2 = 0.957$, Adjusted $R^2 = 0.929$, $F(7, 11) = 34.810$, $Prob > F < 0.001$, Root MSE = 3.472.

For the total score model (Table 5), significant predictors included *AI Attitude*, chatbot usage, and task count. The model demonstrated a high R-squared value, indicating strong explanatory power. Participants with more favourable AI attitudes and those using the chatbot achieved higher total scores. Task count also had a positive effect, as completing more tasks naturally contributed to higher total scores. Other factors, such as experience, education, and AI familiarity, did not significantly impact total scores.

For the average score per task model (Table 6), *AI Attitude*, chatbot usage, and task count again emerged as significant predictors. The model demonstrated a similarly strong fit. Participants with favourable AI attitudes and those using the chatbot achieved higher average task scores, emphasising the assistant's contribution to task quality. However, task count was negatively associated with the average score per task. While completing more tasks boosted total scores, it appeared to lower average quality, likely due to time constraints or fatigue.

Both analyses confirmed that *AI Attitude* and chatbot usage consistently predicted better performance across metrics. Specifically, the chatbot contributed to improvements in both total scores and average scores per task. Task count showed a dual effect: it increased total scores but reduced average task quality. This suggests a trade-off between the number of responses completed and their individual quality.

**Table 6.** Regression results for average score per task.

| Variable | Coefficient | Std. err. | [95% CI lower] | [95% CI upper] |
|---|---|---|---|---|
| *AI Efficiency* | −0.082 | 0.048 | −0.189 | 0.024 |
| *AI Attitude* | 0.286** | 0.096 | 0.076 | 0.497 |
| Experience | 0.169 | 0.095 | −0.041 | 0.379 |
| Education | 0.139 | 0.126 | −0.140 | 0.417 |
| AI familiarity | −0.260* | 0.133 | −0.552 | 0.033 |
| Chatbot usage | 1.088*** | 0.257 | 0.522 | 1.654 |
| Task count | −0.062** | 0.023 | −0.113 | −0.012 |
| Intercept | 1.084** | 0.444 | 0.107 | 2.062 |

Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.Model Fit Statistics: $R^2 = 0.827$, Adjusted $R^2 = 0.718$, $F(7, 11) = 7.530$, $Prob > F = 0.002$, Root MSE = 0.355.

**Table 7.** Regression results for the subgroup analysis of experience levels with the focus on the interaction of chatbot usage and experience group.

| | Total score | | Average score per task | |
|---|---|---|---|---|
| Variable | Coefficient | Std. err. | Coefficient | Std. err. |
| Chatbot usage[a] | 13.170*** | 2.839 | 1.687*** | 0.305 |
| Experience group[b] | 9.911*** | 2.949 | 0.935** | 0.317 |
| Interaction term | −8.629** | 3.552 | −0.876** | 0.381 |
| *AI Efficiency* | −0.503 | 0.396 | −0.088* | 0.042 |
| *AI Attitude* | 2.243** | 0.778 | 0.264*** | 0.083 |
| Education | 1.478 | 1.005 | 0.055 | 0.108 |
| AI familiarity | −2.487** | 1.029 | −0.327** | 0.110 |
| Task count | 2.249*** | 0.194 | −0.048** | 0.021 |
| Constant | −15.624*** | 2.828 | 1.188*** | 0.304 |
| $R^2$ | 0.973 | | 0.883 | |
| Adjusted $R^2$ | 0.952 | | 0.789 | |
| $Prob > F$ | 0.000 | | 0.001 | |
| Root MSE | 2.859 | | 0.307 | |

Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. [a]Chatbot usage is a binary variable with 0 = non-users and 1 = users. [b]Experience group is also binary with 0 = less experienced and 1 = more.

### 5.3. Subgroup analysis for experience level

To assess whether the chatbot's effectiveness varied with participant experience, we conducted a regression analysis including an interaction term between experience level and chatbot use. Participants were divided into two groups: 'less experienced' (< 3 years) and 'more experienced' (≥ 3years), to examine whether experience moderated the chatbot's impact.

Chatbot usage significantly predicted higher total scores and average scores per task, as shown in Table 7. However, the interaction between experience and chatbot usage showed that less experienced participants gained greater benefit from chatbot support. This suggests that the chatbot's structured guidance particularly supported less experienced participants, enhancing their problem-solving efficiency. Controlling for factors such as *AI Attitude* and task count confirmed that participants with a favourable AI attitude made more effective use of the chatbot.

In summary, the chatbot benefitted all participants but was especially valuable for those early in their careers, reinforcing its role as a support tool for less experienced professionals.

**Table 8.** Regression results for the subgroup analysis of AI familiarity, showing the results for participants with low AI familiarity.

| | Total score | | Average score per task | |
|---|---|---|---|---|
| Variable | Coefficient | Std. err. | Coefficient | Std. err. |
| *AI Efficiency* | −0.204 | 0.474 | −0.036 | 0.045 |
| *AI Attitude* | 2.617** | 1.098 | 0.254** | 0.104 |
| Experience | 1.524 | 0.909 | 0.154 | 0.086 |
| Education | 2.889** | 1.193 | 0.243* | 0.113 |
| Chatbot usage | 6.261** | 2.143 | 0.951*** | 0.203 |
| Task count | 2.056*** | 0.213 | −0.070*** | 0.020 |
| Constant | −19.515*** | 3.958 | 0.729* | 0.375 |
| $R^2$ | 0.965 | | 0.887 | |
| Adjusted $R^2$ | 0.942 | | 0.812 | |
| $Prob > F$ | 0.000 | | 0.001 | |
| Root MSE | 3.200 | | 0.303 | |

Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### 5.4. Subgroup analysis for low AI familiarity

We conducted a separate analysis focussing on participants with low AI familiarity, specifically those who reported 'Not familiar at All' or 'Somewhat familiar' with AI in response to Q4 (Appendix 1). This analysis examined whether the performance predictors from the full sample also applied to this subgroup and identified factors specifically affecting less AI-familiar participants.

In the total score model (Table 8, left panel), significant predictors included *AI Attitude*, education, chatbot usage, and task count. Participants with a positive AI attitude and higher education performed better, suggesting these factors mitigated the effects of low AI familiarity. Chatbot usage also significantly predicted better performance, emphasising its value for participants less familiar with AI. Chatbot usage also remained a strong predictor in the average score per task model (Table 8, right panel). Conversely, task count had a slight negative association, likely reflecting trade-offs between task quantity and quality.

Participants with low AI familiarity showed significantly improved performance when using the chatbot, with positive attitudes toward AI amplifying this effect. Higher education levels helped mitigate the impact of limited AI familiarity by supporting task comprehension and execution.

### 5.5. User feedback from post-experiment survey

After the experiment, participants completed a post-survey assessing its *ease of use*, *interface intuitiveness*, *satisfaction with answers*, and *satisfaction with sources* (Figure 4). Overall responses were positive, though they varied across categories: *ease of use* ratings displayed

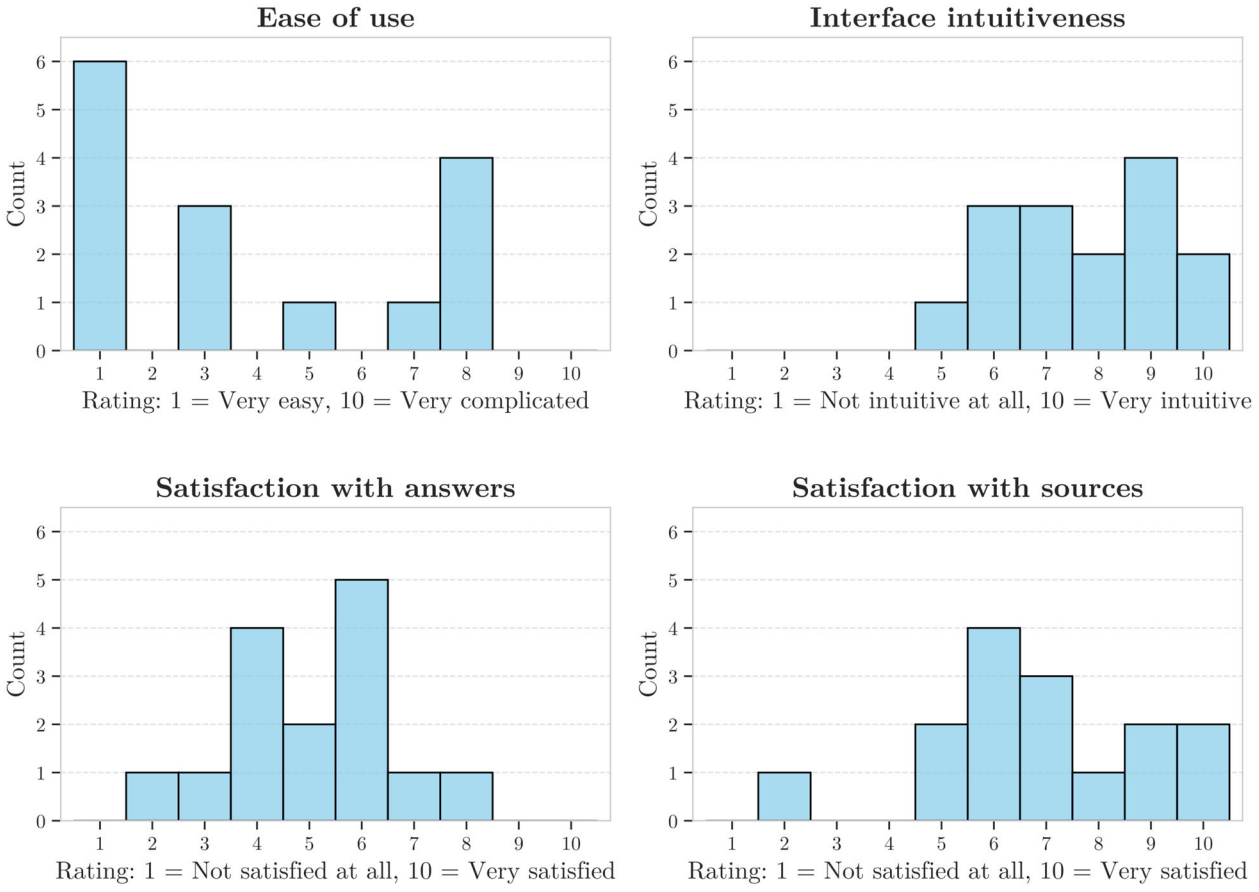



**Figure 4.** Participant satisfaction ratings for the chatbot condition ($N = 15$).

a bimodal distribution, reflecting mixed user experiences, while *interface intuitiveness* consistently received high scores (7–10). Moderate scores for *satisfaction with answers* and a broader range for *satisfaction with sources* indicate that while participants valued the chatbot's utility, they also recognised areas needing improvement.

Correlations among satisfaction measures were generally weak, except for a moderate positive relationship between *ease of use* and *interface intuitiveness* (Spearman's $\rho = 0.373$), suggesting that users who found the chatbot easy to use also perceived its interface as intuitive.

Among respondents, 42.11% ranked answer accuracy as the most valuable feature, followed by ease of documentation search (31.58%) and interface usability (26.32%). Participants suggested expanding the knowledge base and improving response speed as priority areas for enhancement, while fewer comments focussed on accuracy and interface adjustments. Open-ended feedback highlighted challenges with language consistency and the handling of project-specific terminology. Based on these observations, participants suggested displaying source excerpts for verification, aligning responses with project terminology, and adding filtering options for more relevant searches.

Participants saw potential in using the chatbot for troubleshooting and quick information retrieval, especially for time-sensitive or fieldwork tasks. Overall, participants praised the chatbot's accuracy and searchability, while recommending improvements in speed, content breadth, and interface consistency. These insights suggest the potential of the chatbot as a valuable support tool for troubleshooting and task guidance in industrial environments.

## 6. Discussion

The primary goal of this study was to assess an AI-enabled troubleshooting assistant designed to improve knowledge accessibility and task performance, with user satisfaction evaluated as an outcome of its use. Drawing on data from a controlled experiment and post-experiment survey, the study offers empirical insights into the assistant's effects on task quality and user experience, with practical implications for deploying generative AI in industrial commissioning settings. The findings are consistent with broader industry developments in AI-driven knowledge management and the increasing emphasis on user-centred AI applications.

### 6.1. The effect of the AI assistant on performance

Participants with access to the chatbot achieved higher total and average scores, indicating that the assistant improved both response accuracy and contextual relevance. This result is consistent with prior evaluations of similar AI tools, such as the ChatMES by Do and Jeong (2022), which streamlined data access by simplifying interactions with complex systems. However, the lack of a significant difference in task count between chatbot users and non-users suggests that the chatbot's primary contribution lies in enhancing output quality rather than speed. This distinction is important, as task-oriented AI systems are typically adopted to enhance the quality of outputs rather than to maximise throughput (Dell'Acqua et al. 2023).

Prior research emphasises that task-specific AI tools, like the one evaluated in this study, can enhance productivity without overwhelming users through automation (Brynjolfsson, Li, and Raymond 2025). This study extends that perspective by demonstrating that a chatbot can improve accuracy in real-world commissioning scenarios. Framing AI implementation around 'quality over speed' may offer a more sustainable path to improved user satisfaction and effective troubleshooting in industrial contexts.

### 6.2. Role of user attitude and familiarity with AI

User attitudes toward AI and prior familiarity with the technology emerged as significant predictors of task performance. Participants with more favourable attitudes toward AI achieved higher task scores, suggesting that openness to the technology may facilitate more effective engagement with the assistant. This aligns with findings by Kernan Freire, Wang, and Niforatos (2024), who argue that positive AI experiences can reinforce favourable attitudes toward its broader applications. For participants with limited AI familiarity, educational background played a compensatory role, highlighting how foundational knowledge can facilitate effective engagement with AI tools. Formal education may provide users with limited AI familiarity and the transferable skills needed to use such tools effectively, reinforcing findings by Wang et al. (2023) on the value of targeted training for novice users.

Subgroup analyses indicated that participants with lower AI familiarity derived greater benefit from chatbot use than their more experienced counterparts, likely due to the structured, accessible nature of the support it provided. This finding supports insights by Brynjolfsson, Li, and Raymond (2025), who emphasise that generative AI can help close skill and experience gaps by delivering structured guidance, especially in heterogeneous user populations. The inclusion of onboarding programs or targeted tutorials may further enhance outcomes, as advocated by Jarrahi et al. (2023), who stress the importance of AI literacy to ensure effective tool integration.

### 6.3. User satisfaction

Feedback from the post-experiment survey indicated overall user satisfaction, particularly with interface intuitiveness and response accuracy. A positive correlation between *ease of use* and *interface intuitiveness* suggests that intuitive design contributed meaningfully to user experience. This aligns with findings from Casillo et al. (2020), who report that intuitive AI interfaces are key drivers of user satisfaction and system adoption. Participants also valued the chatbot's accuracy and efficiency in searching documentation, highlighting the importance of reliable information retrieval for supporting complex troubleshooting. This mirrors the conclusions of Gröger, Schwarz, and Mitschang (2014), who emphasise the role of structured knowledge access in industrial knowledge management.

The findings suggest that AI tools in commissioning contexts should prioritise accuracy, accessibility, and ease of use, especially for technicians who rely on timely and dependable information. Participant suggestions to expand the chatbot's knowledge base and improve response speed align with broader industry trends, such as expanding domain coverage and improving system responsiveness for effective AI deployment. For example, Zheng et al. (2024) emphasise the importance of tailoring information delivery to users' needs to ensure the adaptability of digital assistants in industrial settings. Implementing such improvements could further enhance the chatbot's utility in complex engineering environments.

### 6.4. AI can bridge the experience gap

The chatbot's benefits were significantly more pronounced among less experienced participants, suggesting that it proved especially effective for individuals in the earlier stages of their careers. This observation aligns with findings by Brynjolfsson, Li, and Raymond (2025) and Kernan Freire et al. (2023), who emphasise the role of AI in capturing tacit knowledge from experienced workers and making it accessible to less experienced employees. The assistant appears to bridge knowledge gaps in real-time by offering structured guidance, consistent with findings by Penica et al. (2023), who show that AI tools can support continuous learning by transferring

experience-based insights to novice employees. Integrating AI-driven tools into commissioning workflows could therefore promote skill development among less experienced technicians, ultimately contributing to a more resilient and capable workforce. However, organisations should remain cautious about the potential risks of AI over-reliance. Recent research on the educational impacts of generative AI suggests that users may become dependent on these tools, resulting in diminished personal learning gains compared to those working without AI support (Bastani et al. 2024).

### 6.5. Contributions to the existing literature

This study presents one of the first empirical evaluations of an AI-enabled troubleshooting assistant, addressing a critical gap in the literature by systematically assessing the role of AI in real-time problem-solving. Although prior studies have explored AI-driven knowledge management, predictive maintenance, and automation in manufacturing (Fosso Wamba et al. 2024; Kernan Freire, Wang, and Niforatos 2024; Rai et al. 2021), the specific role of AI in supporting direct problem resolution in commissioning workflows remains underexplored. This study positions AI troubleshooting assistants as a distinct category within AI-driven industrial support systems, demonstrating their effectiveness in enhancing task execution in complex operational environments.

An additional contribution is the empirical validation of RAG in a troubleshooting context, demonstrating that AI systems tailored for technical problem-solving can deliver context-aware and accurate responses that enhance task performance. Unlike general-purpose chatbots such as ChatMES and OEEAI bot (Do and Jeong 2022; Loh et al. 2023), which emphasise structured data retrieval, the assistant developed in this study was designed for adaptive troubleshooting, thereby addressing a critical gap in practical AI applications for industrial problem-solving.

## 7. Conclusions and outlook

We developed a troubleshooting AI assistant and evaluated it using realistic scenarios from the train manufacturing context. The results demonstrate that the assistant significantly improves task quality and supports user satisfaction in commissioning tasks. The positive impact of the chatbot on both performance and satisfaction highlights its potential as a valuable support tool, especially for technicians with limited experience or AI familiarity. These findings contribute empirical evidence to the growing literature on generative AI in industrial applications.

Consistent with existing literature on the productivity benefits of AI in industrial settings, this study adds nuance by highlighting the chatbot's potential as a quality-enhancing tool for knowledge-intensive troubleshooting, particularly for less experienced workers (Brynjolfsson, Li, and Raymond 2025; Dell'Acqua et al. 2023). While previous studies have primarily emphasised productivity gains, this study adds important nuance by showing that the assistant's principal contribution lies in quality improvements rather than task speed. Contrary to findings by Choi and Schwarcz (2023), which suggest that AI support may be less effective in complex, high-stakes tasks, this study demonstrates that tailored AI solutions can remain effective in such settings when designed with a focus on accuracy and usability.

### 7.1. Identified improvement areas for future AI assistants

Although the results of AI-assisted train commissioning are promising, several opportunities remain to enhance the AI assistant's effectiveness. Specific improvement areas include expanding the knowledge base, increasing response speed, and aligning terminology with project-specific language. These enhancements would directly respond to user feedback on the limitations of the chatbot. Participants suggested including source excerpts or images to verify answers, complementing the current system of document references. In line with recommendations by Wellsandt et al. (2021) for hands-free, real-time support systems, such enhancement would strengthen the chatbot's utility and reliability in industrial applications. Similarly, aligning the chatbot's responses with specialised terminology would address the need for industry-specific language adaptation, as discussed by Kernan Freire, Foosherian, et al. (2023) and Yang et al. (2023). As Zheng et al. (2024) emphasise, adapting AI systems to the specific requirements of industrial users is essential for effective implementation. Addressing these improvement areas could elevate the chatbot from a supportive assistant to an indispensable component of the commissioning workflow, enhancing its relevance and usability for field technicians.

Feedback from TrainCo employees suggests that the chatbot could be extended beyond troubleshooting to offer real-time support in the field, particularly in situations where experienced staff are unavailable. This aligns with the assistant's original goal of reducing dependence on experts for routine knowledge and troubleshooting tasks. Specific suggestions include improving the chatbot's accuracy by initially restricting searches to subsystem-specific documents, with the option to expand results if needed. A 'chatbot light' version that functions

purely as an advanced search tool could also be beneficial, reducing response times and operational costs by bypassing complex AI processing.

Another suggested enhancement is the integration of an error database. Enabling the chatbot to store unresolved issues alongside their eventual solutions could support the development of a continuously expanding knowledge repository. Such a system could streamline the resolution of non-hardware issues and facilitate faster troubleshooting for future cases. Adapting the chatbot for departments such as warranty or customer support could broaden its applicability across operational domains, enabling efficient retrieval of past cases and quick access to project-specific guidance.

Further improvement could focus on optimising the chatbot's language comprehension, particularly regarding project-specific terminology, and integrating user feedback to support continuous refinement. Future research may also assess the assistant's applicability in other contexts, such as warranty or customer support, to evaluate its broader utility across departments. Collectively, these directions will support the responsible expansion and refinement of the chatbot as a valuable knowledge tool in technical settings.

### 7.2. Research limitations

While this study offers valuable insights into the effectiveness of an AI-enabled troubleshooting assistant, several limitations warrant consideration.

First, the sample size was relatively small, with only 19 participants completing the experiment. This constraint may limit the generalizability of the findings, as a larger and more diverse participant pool could yield different results, particularly with respect to AI familiarity and differences in technical expertise. Future research should expand the sample size and include participants from additional industrial contexts to evaluate the broader applicability of these findings.

Second, although the study was conducted in a field setting, it did not take place in a live production environment, and therefore may not fully capture the complexities and pressures of real-world commissioning. Typical fieldwork factors – such as environmental distractions, variable task complexity, workplace incentives, and resource constraints – were not replicated in this controlled setting. While the controlled field environment enabled a focussed evaluation of chatbot performance, future research should pursue longitudinal studies in live production contexts to assess the long-term effectiveness of the assistant across varied conditions. Moreover, our experimental design does not capture how interaction with the AI assistant evolves over time or how it affects learning, skill development, and job satisfaction. A critical area for future investigation concerns the influence of the chatbot on users' independent troubleshooting skills and their trust in AI-driven responses, especially in high-stakes scenarios.

Third, this study evaluated the chatbot using a predefined set of task-based scenarios, which may not reflect the full spectrum of challenges technicians face in real-world settings. As a result, the assistant's knowledge base and response accuracy require further evaluation against more complex or unpredictable problems encountered in practice. Moreover, the study relied on a pre-configured knowledge base, which could limit the adaptability of the chatbot in handling unique or evolving technical challenges that require real-time information or updates. Future iterations of the assistant could incorporate adaptive learning mechanisms by integrating ongoing feedback from technicians.

Fourth, while the study focussed on task completion and response accuracy, it did not evaluate other critical outcomes, such as the long-term impact on technician learning or skill retention. While immediate performance metrics are essential, understanding how AI assistance influences users' problem-solving autonomy and knowledge retention over time would offer a more holistic perspective on its benefits and limitations.

Fifth, although participants expressed general satisfaction with the chatbot, the study did not examine the role of trust in AI in shaping engagement and performance. Prior research identifies trust as a key factor in AI adoption and effectiveness in industrial settings, influencing chatbot usage, response reliance, and overall user satisfaction (Kyung and Kwon 2022). Future research should investigate how varying levels of trust affect these outcomes, particularly among users with limited AI familiarity.

Finally, a further limitation of our study concerns the availability of generative AI tools for TrainCo during the experiment. As the field of LLMs evolves rapidly, newer models with enhanced capabilities continue to emerge. At the time of this study, the most advanced model available to TrainCo was GPT-3.5, which may be viewed in hindsight as a limitation relative to more recent models. For example, employing GPT-4 instead of GPT-3.5 would likely improve accuracy, contextual awareness, and response consistency, as demonstrated in a recent study comparing the models in university-level coding tasks (Yeadon, Peach, and Testrow 2024). That study found GPT-4 significantly outperformed GPT-3.5, especially in response to prompt engineering, suggesting that a GPT-4-powered assistant would deliver more intuitive and reliable support for commissioning technicians. However, since this study compared chatbot users

to non-users – rather than evaluating different foundation models – the observed performance differences are likely to become even more pronounced as newer models are integrated into AI assistants.

## Notes

1. https://docs.llamaindex.ai/en/stable/
2. https://flask.palletsprojects.com/en/3.0.x/
3. Cronbach's alpha is a widely accepted measure of internal consistency, with values typically above 0.70 indicating acceptable reliability. Pflughoeft et al. (2003) reported alpha values between 0.73 and 0.92, supporting its use even with moderate sample sizes. While alpha may underestimate reliability, it remains a standard metric, and values above 0.70 are considered sufficient, even for small samples.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Notes on contributors



*Nils Löwhagen* is a Research Associate at the Chair of Production and Operations Management, ETH Zurich, Switzerland. His research focuses on the application of quantum technologies in operations management, with an emphasis on exploratory approaches that aim to contribute an empirically grounded understanding to the field. His early work included projects in process mining and AI applications.



*Philippe Schwendener* is Head of Commissioning Engineering at Stadler Rheintal in St. Margrethen, Switzerland. In his role, he focuses on improving processes in train commissioning through optimization, automation, and digitalization. He has contributed to several international rail projects in Azerbaijan, Sweden, and the United Kingdom, where he served as subproject manager for software. Philippe holds a master's degree in Electrical Engineering and Information Technology, as well as a Master of Advanced Studies in Management, Technology, and Economics, both from ETH Zurich.



*Torbjørn Netland* is Full Professor of Production and Operations Management at ETH Zurich. Switzerland. Prof. Netland is a Fellow of the European Academy of Industrial Management, a Full Member of IFIP WG5.7, and a Member of the World Economic Forum's Global Future Councils. His research spans industrial engineering and operations management, with a focus on how manufacturers can achieve and sustain operational excellence. He is a co-author of Introduction to Manufacturing: An Industrial Engineering and Management Perspective (Routledge, 2022).

## Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

## ORCID

*Nils Löwhagen* http://orcid.org/0009-0008-0352-4711
*Torbjørn Netland* http://orcid.org/0000-0001-7382-1051

## References

Bastani, Hamsa, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. 2024. "Generative AI Can Harm Learning." *SSRN Electronic Journal*. The Wharton School Research Paper. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4895486.

Berg, Justin M., Manav Raj, and Robert Seamans. 2023. "Capturing Value from Artificial Intelligence." *Academy of Management Discoveries* 9 (4): 424–428. https://doi.org/10.5465/amd.2023.0106.

Brown, Tom B., Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. December, 2020. "Language Models Are Few-Shot Learners." In *Advances in Neural Information Processing Systems*.

Brynjolfsson, Erik, Danielle Li, and Lindsey Raymond. 2025. "Generative AI at Work." *The Quarterly Journal of Economics* 140 (2): 889–942. https://doi.org/10.1093/qje/qjae044

Casillo, Mario, Francesco Colace, Loretta Fabbri, Marco Lombardi, Alessandra Romano, and Domenico Santaniello. 2020. "Chatbot in Industry 4.0: An Approach for Training New Employees." In *Proceedings of 2020 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)*, Takamatsu, Japan, 371–376.

Choi, Jonathan H., and Daniel B. Schwarcz. 2023. "AI Assistance in Legal Analysis: An Empirical Study." *SSRN Electronic Journal, 73 Journal of Legal Education (forthcoming, 2024)*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4539836#.

Choi, Tsan Ming, Subodha Kumar, Xiaohang Yue, and Hau Ling Chan. 2022. "Disruptive Technologies and Operations Management in the Industry 4.0 Era and beyond." *Production and Operations Management* 31 (1): 9–31. https://doi.org/10.1111/poms.13622.

Ciancio, Vincent, Lazhar Homri, Jean Yves Dantan, and Ali Siadat. 2024. "Development of a Flexible Data Management System, to Implement Predictive Maintenance in the Industry 4.0 Context." *International Journal of Production Research* 62 (6): 2255–2271. https://doi.org/10.1080/00207543.2023.2217293.

Colabianchi, Silvia, Andrea Tedeschi, and Francesco Costantino. 2023. "Human-Technology Integration with Industrial Conversational Agents: A Conceptual Architecture and a Taxonomy for Manufacturing." *Journal of Industrial Information Integration* 35:100510. https://doi.org/10.1016/j.jii.2023.100510.

Dell'Acqua, Fabrizio, Edward McFowland III, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Krayer, François Candelon, and Karim R. Lakhani. 2023. "Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality." *SSRN Electronic*

*Journal*. Harvard Business School. Working Paper 24-013. https://dx.doi.org/10.2139/ssrn.4573321.

Do, Sanghoon, and Jongpil Jeong. 2022. "Design and Implementation of RPA Based ChatMES System Architecture for Smart Manufacturing." *WSEAS Transactions on Computer Research* 10:88–92. https://doi.org/10.37394/232018.

Dolgui, Alexandre, and Dmitry Ivanov. 2025. "Internet of Behaviors: Conceptual Model, Practical and Theoretical Implications for Supply Chain and Operations Management." *International Journal of Production Research* 63 (1): 1–8. https://doi.org/10.1080/00207543.2024.2372008.

Farooqui, Ashfaq, Kristofer Bengtsson, Petter Falkman, and Martin Fabian. 2020. "Towards Data-Driven Approaches in Manufacturing: An Architecture to Collect Sequences of Operations." *International Journal of Production Research* 58 (16): 4947–4963. https://doi.org/10.1080/00207543.2020.1735660.

Feuerriegel, Stefan, Jochen Hartmann, Christian Janiesch, and Patrick Zschech. 2024. "Generative AI." *Business and Information Systems Engineering* 66 (1): 111–126. https://doi.org/10.1007/s12599-023-00834-7.

Fosso Wamba, Samuel, Cameron Guthrie, Maciel M. Queiroz, and Stefan Minner. 2024. "ChatGPT and Generative Artificial Intelligence: An Exploratory Study of Key Benefits and Challenges in Operations and Supply Chain Management." *International Journal of Production Research* 62 (16): 5676–5696. https://doi.org/10.1080/00207543.2023.2294116.

Kernan Freire, Samuel, Mina Foosherian, Chaofan Wang, and Evangelos Niforatos. 2023. "Harnessing Large Language Models for Cognitive Assistants in Factories." In *Proceedings of the 5th International Conference on Conversational User Interfaces, CUI 2023*, Eindhoven, Netherlands, 44.

Kernan Freire, Samuel, Chaofan Wang, and Evangelos Niforatos. 2024. "Conversational Assistants in Knowledge-Intensive Contexts: An Evaluation of LLM- versus Intent-Based Systems." Vol. 1, 2. arXiv:2402.04955v2.

Kernan Freire, Samuel, Chaofan Wang, Santiago Ruiz-Arenas, and Evangelos Niforatos. 2023. "Tacit Knowledge Elicitation for Shop-Floor Workers with an Intelligent Assistant." In *Conference on Human Factors in Computing Systems – Proceedings*, Hamburg, Germany.

Gao, Tianyu, Adam Fisch, and Danqi Chen. 2021. "Making Pre-trained Language Models Better Few-shot Learners." In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing* (Volume 1: Long Papers), 3816–3830, Online. Association for Computational Linguistics.

Gao, Yunfan, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2023. "Retrieval-Augmented Generation for Large Language Models: A Survey." arXiv:2312.10997v5.

Gilbert, Stephen, Jakob Nikolas Kather, and Aidan Hogan. 2024. "Augmented Non-hallucinating Large Language Models as Medical Information Curators." *npj Digital Medicine* 7 (1): 1–5. https://doi.org/10.1038/s41746-023-00987-5.

Gröger, Christoph, Holger Schwarz, and Bernhard Mitschang. 2014. "The Manufacturing Knowledge Repository Consolidating Knowledge to Enable Holistic Process Knowledge Management in Manufacturing." In *14th International Conference on Enterprise Information Systems (ICEIS)*, Lisbon, Portugal, 4.

Hoerner, Lorenz, Markus Schamberger, and Freimut Bodendorf. 2023. "Using Tacit Expert Knowledge to Support Shop-Floor Operators through a Knowledge-Based Assistance System." *Computer Supported Cooperative Work: CSCW: An International Journal* 32 (1): 55–91. https://doi.org/10.1007/s10606-022-09445-4.

Jackson, Ilya, Dmitry Ivanov, Alexandre Dolgui, and Jafar Namdar. 2024. "Generative Artificial Intelligence in Supply Chain and Operations Management: A Capability-Based Framework for Analysis and Implementation." *International Journal of Production Research* 62 (17): 6120–6145. https://doi.org/10.1080/00207543.2024.2309309.

Jarrahi, Mohammad Hossein, David Askay, Ali Eshraghi, and Preston Smith. 2023. "Artificial Intelligence and Knowledge Management: A Partnership between Human and AI." *Business Horizons* 66 (1): 87–99. https://doi.org/10.1016/j.bushor.2022.03.002.

Johnson, T. L., S. R. Fletcher, W. Baker, and R. L. Charles. 2019. "How and why We Need to Capture Tacit Knowledge in Manufacturing: Case Studies of Visual Inspection." *Applied Ergonomics* 74:1–9. https://doi.org/10.1016/j.apergo.2018.07.016.

Kassner, Laura, Pascal Hirmer, Matthias Wieland, Frank Steimle, Jan Königsberger, and Bernhard Mitschang. January, 2017. "The Social Factory: Connecting People, Machines and Data in Manufacturing for Context-Aware Exception Escalation." In *Proceedings of the Annual Hawaii International Conference on System Sciences*, Waikoloa, Hawaii, United States of America, 1673–1682.

Kernan Freire, Samuel, C. Wang, M. Foosherian, S. Wellsandt, S. Ruiz-Arenas, and E. Niforatos. 2024. "Knowledge Sharing in Manufacturing Using LLM-Powered Tools: User Study and Model Benchmarking." *Frontiers in artificial intelligence* 7: 1293084. https://doi.org/10.3389/frai.2024.1293084.

Kyung, N., and H. E. Kwon. 2022. "Rationally Trust, but Emotionally? The Roles of Cognitive and Affective Trust in Laypeople's Acceptance of AI for Preventive Care Operations." *Production and Operations Management*, 1–20.

Lewis, Patrick, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, et al. December, 2020. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." In *Advances in Neural Information Processing Systems*.

Loh, Siu Hong, Vincent Jascha Jeuk, Jia Jia Sim, and Kim Ho Yeap. 2023. "Implementation of Artificial Intelligence Chatbot in Semiconductor Manufacturing to Optimize Overall Equipment Effectiveness." In *Proceedings – 13th IEEE International Conference on Control System, Computing and Engineering, ICCSCE 2023*, Penang, Malaysia, 220–225.

Lugovoi, Ivan, Dimitrios A. Andritsos, and Claire Senot. 2022. "Novelty and Scope of Process Innovation: The Role of Related and Unrelated Manufacturing Experience." *Production and Operations Management* 31 (10): 3877–3895. https://doi.org/10.1111/poms.13793.

Netland, Torbjørn, Michael Stegmaier, Cesare Primultini, and Omid Maghazei. 2023. "Interactive Mixed Reality Live Streaming Technology in Manufacturing." *Manufacturing Letters* 38:6–10. https://doi.org/10.1016/j.mfglet.2023.08.141.

Ouyang, Long, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, et al. 2022. "Training Language Models to Follow Instructions with Human Feedback." In *Advances in Neural Information Processing Systems* 35, New Orleans, Louisiana, United States of America.

Penica, Mihai, Mangolika Bhattacharya, William O'Brien, Sean McGrath, Martin Hayes, and Eoin O'Connell. 2023. "Adaptable Decision Making Chatbot System: Unlocking Interoperability in Smart Manufacturing." *Proceedings – 2023 International Conference on Computing, Electronics and Communications Engineering, ICCECE 2023*, Swansea, United Kingdom, 23–29.

Pflughoeft, Kurt A., K. Ramamurthy, Ehsan S. Soofi, Masoud Yasai-Ardekani, and Fatemeh Zahedi. 2003. "Multiple Conceptualizations of Small Business Web Use and Benefit*." *Decision Sciences* 34 (3): 467–512. https://doi.org/10.1111/deci.2003.34.issue-3.

Rai, Rahul, Manoj Kumar Tiwari, Dmitry Ivanov, and Alexandre Dolgui. 2021. "Machine Learning in Manufacturing and Industry 4.0 Applications." *International Journal of Production Research* 59 (16): 4773–4778. https://doi.org/10.1080/00207543.2021.1956675.

Senoner, Julian, Torbjørn Netland, and Stefan Feuerriegel. 2021. "Using Explainable Artificial Intelligence to Improve Process Quality: Evidence from Semiconductor Manufacturing." *Management Science* 68 (8): 5704–5723. https://doi.org/10.1287/mnsc.2021.4190.

Telner, Jason. 2021. "Chatbot User Experience: Speed and Content Are King." In *Advances in Artificial Intelligence, Software and Systems Engineerings*, Vol. 271, 47–54. Deutschland GmbH: Springer Science and Business Media. https://doi.org/10.1007/978-3-030-80624-8.

Trappey, Amy J. C., Charles V. Trappey, Min Hua Chao, and Chun Ting Wu. 2022. "VR-enabled Engineering Consultation Chatbot for Integrated and Intelligent Manufacturing Services." *Journal of Industrial Information Integration* 26:100331. https://doi.org/10.1016/j.jii.2022.100331.

von Wolff, Raphael Meyer, Sebastian Hobert, and Matthias Schumann. January, 2019. "How May I Help You? – State of the Art and Open Research Questions for Chatbots at the Digital Workplace." In *Proceedings of the Annual Hawaii International Conference on System Sciences*, Maui, Hawaii, United States of America, 95–104.

Wang, Lingli, Ni Huang, Yili Hong, Luning Liu, Xunhua Guo, and Guoqing Chen. 2023. "Voice-Based AI in Call Center Customer Service: A Natural Field Experiment." *Production and Operations Management* 32 (4): 1002–1018. https://doi.org/10.1111/poms.13953.

Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi Quoc, V. Le, and Denny Zhou. 2023. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." arXiv:2201.11903.

Wellsandt, Stefan, Konstantin Klein, Karl Hribernik, Marco Lewandowski, Alexandros Bousdekis, Gregoris Mentzas, and Klaus Dieter Thoben. 2021. "Towards Using Digital Intelligent Assistants to Put Humans in the Loop of Predictive Maintenance Systems." *IFAC-PapersOnLine* 54 (1): 49–54. https://doi.org/10.1016/j.ifacol.2021.08.005.

Yang, Fangkai, Pu Zhao, Zezhong Wang, Lu Wang, Jue Zhang, Mohit Garg, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. 2023. "Empower Large Language Model to Perform Better on Industrial Domain-Specific Question Answering." In *EMNLP 2023 – 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Industry Track*, Singapore, 294–312.

Yeadon, Will, Alex Peach, and Craig P. Testrow. 2024. "A Comparison of Human, GPT-3.5, and GPT-4 Performance in a University-Level Coding Course." *Scientific Reports* 14 (1): 1–9. https://doi.org/10.1038/s41598-024-73634-y.

Zalake, Niranjan, and Gautam Naik. 2019. "Generative Chat Bot Implementation Using Deep Recurrent Neural Networks and Natural Language Understanding." In *Proceedings 2019: Conference on Technologies for Future Cities (CTFC)*, Navi Mumbai, Maharashtra, India. http://doi.org/10.2139/ssrn.3362123.

Zheng, Ting, Eric H. Grosse, Stefan Morana, and Christoph H. Glock. 2024. "A Review of Digital Assistants in Production and Logistics: Applications, Benefits, and Challenges." *International Journal of Production Research* 62 (21): 8022–8048.

Zimmer, Melanie, Ali Al-Yacoub, Pedro Ferreira, and Niels Lohse. 2020. "Towards Human-Chatbot Interaction: A Virtual Assistant for the Ramp-up Process." In *UKRAS20 Conference: 'Robots Into the Real World' Proceedings*, Lincoln, United Kingdom (virtual), Vol. 3, 108–110.

# Appendices

## Appendix 1. Pre-Experiment Survey

### Background Information

Q1. What area do you work in?
☐ Commissioning ☐ Software Engineering
Q2. Years of professional experience in the industry?
☐ Less than 1 year ☐ 1–3 years ☐ 3–5 years ☐ More than 5 years
Q3. Educational qualifications?
☐ Apprenticeship ☐ Bachelor's degree ☐ Master's degree
Q4. Familiarity with AI?
☐ Not familiar at all ☐ Somewhat familiar ☐ Familiar ☐ Very familiar
Q5. Which AI tools have you worked with so far?
*(Open text)*
Q6. Satisfaction with AI tools used:
*1 = Not at all satisfied, 10 = Very satisfied*

### General Attitudes Toward AI

*Rate how strongly you agree with each statement (1 = Strongly disagree, 10 = Strongly agree):*

| Statement | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Q7. AI has a positive impact on society. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q8. AI will create more jobs than it will eliminate. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9. AI will improve the quality of life. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q10. AI poses risks for data protection. *(reverse-coded)* | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q11. AI technologies are trustworthy. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

### AI Efficiency Perception

*To what extent do you agree with the following statements?*

| Statement | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Q14. AI would improve my efficiency at work. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q15. AI would make my work more interesting. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q16. AI would help me make better decisions. | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

### AI Interaction and Use in the Workplace

Q12. How often do you interact with AI in your current role?
☐ Never ☐ Rarely ☐ Sometimes ☐ Often ☐ Always
Q13. Which AI applications would be useful in your role? (multiple answers)
☐ Predictive maintenance ☐ Quality control ☐ Knowledge management
☐ Generative AI ☐ Other: *(specify)*
Q17. I'm worried that AI will replace my job.
☐ Strongly disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly agree
Q18. Concerns about AI implementation? (multiple answers)
☐ Workplace safety ☐ Data protection ☐ Reliability of AI systems
☐ Lack of understanding/training ☐ Ethical aspects ☐ Quality of output ☐ Other: *(specify)*
Q19. How much of your workday is spent searching for information?
*Scale: 1 = 0%, 10 = 100%*
Q20. Any comments or suggestions on AI in your workplace?
*(Open text)*

## Appendix 2. Lab Experiment Tasks

Each task below reflects a realistic troubleshooting scenario from the field. Unless instructed otherwise, respond using bullet points or brief notes.

| No. | Task title | Scenario | Instructions |
|---|---|---|---|
| 1 | Park Function Blocked | You have just replaced the contactor with which the VCU can enable the compressor. After the work has been completed, you want to put the vehicle in park. However, the 'P' button on the IDU is blocked, and when you press the hardware 'P' button, it flashes. | List possible reasons why the parking position cannot be reached. |
| 2 | Impact Protection Fault | Your PLI informs you that the impact protection on vehicle 005 has failed due to a broken contact strip and gives you the order to get the vehicle ready for the test drives in 2 weeks. | Describe in keywords the steps needed to rectify the fault. |
| 3 | GSM-R Phonebook Upload | You need to load the phonebook onto the GSM-R system. | Describe the procedure in bullet points. |
| 4 | ESS Activation Failure | During the SW validation of the HSCB, SW Engineering opened and closed the HSCB several times within a few minutes and now wants to continue testing the Energy Storage System (ESS). They cannot activate the ESS, so they call you as the person responsible for the vehicle. | • What do you tell them is the probable reason why the ESS can no longer be switched on?<br>• How can the problem be solved?<br>• What advice would you give them about what to look for next time they test the HSCB? |
| 5 | TELOC Data Readout | You are asked to read out data from the TELOC after a test drive. | • What material is needed?<br>• How do you connect to the TELOC?<br>• Describe the readout procedure. |
| 6 | Compressor Fault During Maneuvering | You receive the order tomaneuver the vehicle at the depot.Whenyou arrive at the vehicle, you see that the SPLpressure is too low, although the compressor is running, the pressure is not rising. | Describe your approach to maneuver the vehicle despite a defective compressor. |
| 7 | Air Conditioning Restart | After replacing an appliance, you receive the order to put the air conditioning system back into operation. | Describe the procedure in bullet points. |
| 8 | HSCB Not Closing | You are given the task of maneuvering a vehicle that is switched off. After you have switched on the vehicle, you lift the panto, and the vehicle detects high voltage. It is still not possible to close the HSCB. | List the possible causes that prevent the HSCB from closing. |
| 9 | Sliding Step Removal | You are on a test drive and notice that the passenger door's sliding step can no longer be retracted. You decide to remove the sliding step. | Briefly describe the procedure to remove the step. |
| 10 | Prepare Vehicle for Towing | You are on a test drive and have a problem with the power converters. It is impossible to rectify the problem, and you cannot continue driving without the converters. You request a vehicle to tow you away. While waiting for the vehicle, you must prepare the defective vehicle for towing. | Outline steps to prepare the vehicle for towing. |
| 11 | SW-Config Installation | Your PLI gives you the task of loading vehicle 005 with SW-Config SW_10, which was released by SW-Engineering an hour ago. | Describe the procedure in keywords. |
| 12 | Panto Fails to Lift | You are given the task of maneuvering a vehicle that is switched off. After the vehicle has been switched on, you press the button to lift the panto. The panto does not lift despite repeated attempts. | List the possible causes that prevent the panto from lifting. |
| 13 | Wheel Slide Protection Test | You are to performa wheel slide protection (anti-skid) test. | • How is the anti-skid test started?<br>• What conditions are required to start the test?<br>• What is the general test procedure?<br>• How is the valve control handled? |
| 14 | Train Protection Disconnection | During a test drive, a fault occurs in the train protection system (ATS) that cannot be rectified. The decision led to a disconnect in the system. | • Describe in bullet points how the train protection can be disconnected.<br>• Describe the effects on the journey if train protection is disconnected. |
| 15 | Emergency Brake Reset | The VCU has triggered an emergency brake application due to the full traction requirement being detected and a speed of 12 km/h. | How can the emergency braking be reset? |
| 16 | Compressor Control Behavior | During preparation for operation in single traction mode with overhead line voltage applied, you are asked to verify the correct automatic behavior of the compressor system. The vehicle is not in park position, and system pressure needs to be monitored. | Describe the switching on and off behaviour of the compressors depending on the pressure of the supply line in single traction with overhead line voltage and when the vehicle is not in the park position. |
| 17 | Safety Test Activation | The DDS message 'Safety test due' is pending. | • How can you start the 'Safety Test'?<br>• What is checked during the safety test? |
| 18 | Power Converter Does Not Cycle | You are instructed to maneuver a vehicle that is switched off. After you have switched on the vehicle, lifted the panto, and pressed the button to close the main switch, the power converter does not start to cycle. | • List the possible causes for this fault.<br>• Describe the troubleshooting in keywords. |
| 19 | ATS Recommissioning | After replacing a device, you receive the order to put the Train Protection (ATS) back into operation. | Describe the procedure in bullet points. |
| 20 | Cyclical Door Test | You have repaired the door and want to perform the cyclical door test. | How can the test be started? |
| 21 | DAS Update | You are tasked with updating theDriverAdvisory System(DAS) for a test drive. | • What material is required?<br>• Describe the update procedure. |
| 22 | Emergency Brake from TELOC | Due to a speed difference between the TELOC speed and the SPEEDO speed, the TELOC sends an emergency brake command to the SCU. The emergency braking cannot be cancelled by acknowledging it. | How can the emergency brake be reset? |

## Appendix 3. Post-Experiment Survey

This survey captures participants' experience and feedback after using the chatbot tool during the lab experiment. Responses include Likert ratings, multiple-choice selections, and open-ended answers.

Likert-Scale Questions
*Please rate each item on a scale from 1 to 10.*

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Q1. How easy is it to use the chatbot? *(1 = Very easy, 10 = Very complicated)* | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q2. How intuitive do you find the user interface? *(1 = Not intuitive at all, 10 = Very intuitive)* | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3. How satisfied are you with the formulated answers that the chatbot gave to your questions? *(1 = Not satisfied at all, 10 = Very satisfied)* | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4. How satisfied are you with the sources the chatbot provided in response to your questions? *(1 = Not satisfied at all, 10 = Very satisfied)* | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Open Text Responses

Q5. Have you noticed any errors or problems when using the chatbot?
Q6. Please list the errors/problems you encountered.
Q8. What questions or tasks can you imagine asking the chatbot in the future?
Q10. Do you have any suggestions for improving the chatbot?
Q11. In which areas of your day-to-day work could you imagine using the chatbot?
Q12. Further feedback or comments…

Multiple Choice Questions (Select all that apply)
*Q7. Which aspects of the chatbot do you find most useful?*

☐ Quick answering of questions
☐ Easy searching of the project documentation
☐ Accuracy of the answers
☐ User-friendly interface
☐ Indication of sources
☐ Anonymity when asking questions
☐ Other features

*Q9. What improvements would you like to see in the chatbot?*

☐ Better accuracy of answers
☐ Faster response time
☐ Improvement of the user interface
☐ Expansion of the database/knowledge base
☐ Other improvements